CST8116 Pre-Lab Exercise 01 Last Revision 10Sep 24 DHaley

Learning Resources

- Review the PowerPoint slides and Microsoft Word documents in lecture notes for Week 1.
- Complete Hybrid 1 to install Java and the Eclipse IDE.
    - Ask your lab professor for help with software installation in the lab as needed.
- Complete the required reading in Week 1.
    - Joyce Farrell. 2018. Programming Logic & Design Comprehensive. 9th Ed. Cengage Learning.
      - Read Chapter 1
    - Cay Horstmann. 2019. Big Java Early Objects. 7th Ed. Wiley.
      - Read Chapter 1 For eText users, interact with the end-of-chapter questions to practice concepts.

Learning Objectives

1. Practice how to design an algorithm using pseudocode and a flowchart.
2. Practice the basics of Java programming by creating a simple program that outputs text to the console.

General Instruction: You should complete the following tasks before your scheduled lab period.

First : input Sum1 and Sum2
Second : add Sum1 + Sum2 into = Total Sum
Third : Output Total Sum

Task 1 – Problem Solving

A client wants a computer program that will add two numbers and output the result on the screen. Given the numbers 42 and 8, add them, and output the result.

What are the three steps needed, in a general sense, for a computer program to solve the problem?

Task 2 – Pseudocode

Another programmer at work is using pseudocode to plan the logic for a computer program and has asked you to review their work. The program should input 2 numbers, calculate the sum, and output the sum.

Re-write the pseudocode to correct it (correct the sequence of steps) and place your corrected pseudocode into your MS Word document.

```
start
  sum = number1 + number2
  input number1
  output "Enter number 1 "
  Declarations
     num number1
     num number2
     num sum
  output "Sum is: ", sum
  output "Enter number 2 "
  input number2
stop
```

```
Start

Declarations
    num number1
    num number2
    num sum

output "Enter number 1"
Input number 1
output "Enter number 2"
Input number 2

sum = number1 + number2

output "Sum is: ", sum

Stop
```

Task 3 – Testing

Use the following test plan to verify that your corrected pseudocode from Task 2 is correct. In other words, start with 'start' and trace your logic line by line documenting the output. If your output does not match the expected output, re-examine your sequence of steps. Place the completed test plan into your MS Word document.

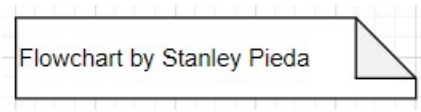
Input	Expected Output	Actual Output	Description
42 8	Enter number 1 Enter number 2 Sum is: 50		Testing the method's main logic, it should ask for two inputs, and then output the sum of the entered numbers.

Task 4 – Flowchart

The company you work for has a contact with a client to provide flowchart(s) as part of the project delivery. Using diagrams.net (See Week 1 lecture notes for handout) convert the provided pseudocode into a flowchart and place the flowchart as an image into your MS Word document.

```
start
  Declarations
     string name
     string message
  output "Please enter name "
  input name
  message = "Hello ", name, " have a great day"
  output message
stop
```

Note: Your name must appear within the flowchart as the author, for example:



Task 5– Java Program

Using Eclipse, create a project named Pre-Lab Exercise 01.

Use the code provided below to create a class named Program, modify the source code so the program outputs your name instead of “student”.

Provide programmer comments in the source code file as per the lecture materials.

Take a screenshot of the Eclipse console showing the result of running the program and place this into your MS Word document.

Save your Java source code file alongside your MS Word document.

```
public class Program {

	public static void main(String[] args) {

		String name = "Student";

		System.out.println("Welcome to CST8116!");

		System.out.print("Java says: Hello ");

		System.out.print(name);

		System.out.println(); // new-line

		System.out.println("Program modified by " + name);

	}

}
```